

● HARD

● GEOMETRY AND TRIGONOMETRY

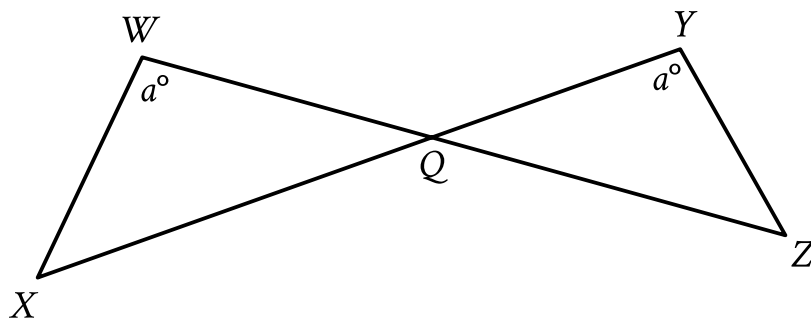
Geometry and Trigonometry

03

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- Triangle WXY has a common vertex with triangle XZY .
- The common vertex is Q .
- For triangle WXY :
 - The measure of angle W is a° .
- For triangle XZY :
 - The measure of angle Y is a° .
- A note indicates the figure is not drawn to scale.

In the figure shown, WZ and XY intersect at point Q , $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of YZ ?

- A. 18
- B. 36
- C. 120
- D. 200

Q2

GRID-IN

GEOMETRY AND TRIGONOMETRY

A rectangular poster has an area of 360 square inches. A copy of the poster is made in which the length and width of the original poster are each increased by 20%. What is the area of the copy, in square inches?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GRID-IN

GEOMETRY AND TRIGONOMETRY

A circle in the xy -plane has a diameter with endpoints $2, 4$ and $2, 14$. An equation of this circle is $x^2 + y^2 - 9^2 = r^2$, where r is a positive constant. What is the value of r ?

STUDENT-PRODUCED RESPONSE

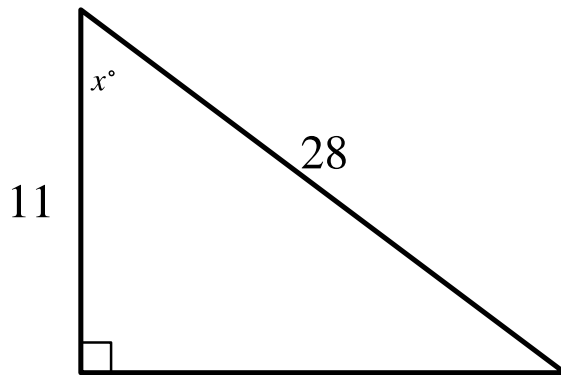
.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The length of the hypotenuse is 28.
- The length of the leg adjacent to the angle with measure x° is 11.
- A note indicates the figure is not drawn to scale.

In the triangle shown, what is the value of $\cos x^\circ$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Triangle XYZ is similar to triangle RST such that X , Y , and Z correspond to R , S , and T , respectively. The measure of $\angle Z$ is 20° and $2XY = RS$. What is the measure of $\angle T$?

- A. 2°
- B. 10°
- C. 20°
- D. 40°

Circle A has a radius of $3n$ and circle B has a radius of $129n$, where n is a positive constant. The area of circle B is how many times the area of circle A ?

- A. 43
- B. 86
- C. 129
- D. 1,849

Point F lies on a unit circle in the xy -plane and has coordinates $1, 0$. Point G is the center of the circle and has coordinates $0, 0$. Point H also lies on the circle and has coordinates $-1, y$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

- A. $\frac{27\pi}{2}$
- B. $\frac{29\pi}{2}$
- C. 24π
- D. 25π

For two acute angles, $\angle Q$ and $\angle R$, $\cos Q = \sin R$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 61$ and $4x + 4$, respectively. What is the value of x ?

- A. 5
- B. 19
- C. 23
- D. 29

Q9

GRID-IN

GEOMETRY AND TRIGONOMETRY

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $9x - 560^\circ$. The sum of the measures of one of the acute angles and three of the obtuse angles is $-18x + w^\circ$. What is the value of w ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The floor of a ballroom has an area of 600 square meters. An architect creates a scale model of the floor of the ballroom, where the length of each side of the model is $\frac{1}{10}$ times the length of the corresponding side of the actual floor of the ballroom. What is the area, in square meters, of the scale model?

- A. 6
- B. 10
- C. 60
- D. 150